

PREHISTORIC VR SIMULATIONS AS EDUCATIONAL CONTENT

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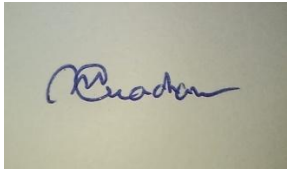
Department of Information Technology

Sri Lanka Institute of Information Technology
Sri Lanka

September 2025

DECLARATION

I declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Abstract

Traditional methods for teaching prehistoric timelines in museums, exhibits, and schools mainly use one-way learning techniques like static charts and text-based materials. Often, these result in low learner engagement and poor memory retention. This report suggests creating a Prehistoric VR Timeline that uses immersive Virtual Reality (VR) environments to present historical content in a more interactive and memorable way. The system aims to improve user involvement and knowledge retention through narration, pre-questions, micro-interactions, and gamification within 3D prehistoric settings. The report highlights the limitations of current static and digital methods, integrates the Visual-Auditory-Kinesthetic (VAK) learning model, and applies story-based and prediction-driven techniques to spark curiosity. The methodology uses a comparative experimental design to measure learning outcomes between traditional timeline users and VR timeline participants. The proposed solution hopes to show that VR-based timelines can change how prehistoric content is delivered, providing a more inclusive and educational alternative for schools, museums, and exhibitions.

Key words – VR, VAK learning model, Prehistory, Timeline

ACKNOWLEDGEMENTS

Thanks to the kind assistance, direction, and contributions of numerous people and organizations, this component was successfully completed. I want to express my sincere gratitude to everyone who helped along the way.

Throughout the entire development process, I am incredibly grateful to my supervisors, Mr. Ishara Gamage and Mr. Nushkan Nizmi, for their invaluable advice, helpful criticism, and unwavering support. Their knowledge and dedication not only helped to guide this work but also improved my abilities as a professional and learner.

I also want to express my profound gratitude to the Sri Lanka Institute of Information Technology's (SLIIT) Department of Information Technology for providing the tools, space, and learning environment that made this component possible.

We would especially like to thank the peers and subject matter experts whose expertise and insights were invaluable throughout the design, implementation, and assessment stages. Their assistance made sure that this work was accurate, useful, and pertinent.

I also want to express my gratitude to everyone who helped and offered feedback during the testing and validation process. Their work enabled me to improve the solution and bring it into line with practical requirements. The combined commitment and assistance of many people and organizations have made this component possible. This accomplishment would not have been possible without the contributions of each and every one of them, for which I am truly grateful.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
VR	Virtual Reality
XR	Extended Reality
VAK	Visual, Auditory, and Kinesthetic (learning model)
SDK	Software Development Kit

1. INTRODUCTION

1.1 Background Literature

Currently, text-heavy static media like banners and infographics are used to teach and communicate prehistoric timelines. They are common in museums, exhibits, and schools, but they lack interactivity and ways for learners to participate in the process. This makes the content less engaging, less attention-grabbing, and less memorable.

With virtual reality (VR) technology, we can place learners in an immersive, spatial world where they can engage with the timeline. We can use all the components of the VAK learning model, along with other methods, to improve engagement, attention, and memory retention.

We can incorporate guided narrations (auditory), high-quality scenes (visual), micro-interactions, and gamifications (kinesthetic) to apply the VAK model. By providing interaction feedback, pre-questions, and progressive revealing through storytelling, we can enhance attention and engagement while reducing cognitive overload.

Although previous research has shown promising results with 360 videos and two-dimensional web application-based timelines, some gaps still exist. Most notably, there is a one-way learning approach and a lack of learning optimizations.

To test this hypothesis, we will use the Mesozoic era's ecosystem evolution milestones. The Mesozoic era consists of three periods: Jurassic, Triassic, and Cretaceous. The common way of presenting this data is through infographics or banners that include images and text, and rarely through videos. VR is ideal for this application because it can create environments rich in prehistoric content. Virtual reality can bring the nonexistent to life and give students access to settings, creatures, and events they would not otherwise experience.

The system will use a "teleport navigation" method to avoid motion sickness, featuring simple controls and a responsive feedback system to ensure a user-friendly experience, even for those with no prior VR experience. Guided narration with user interaction will help maintain user focus, allowing them to learn the content without feeling bored. The proposed system will be tested against traditional mediums to evaluate how well it meets expectations.

1.2 Research Gap

The study of prehistoric timelines has shifted from static charts and posters to web-based interactive timelines and, more recently, experimental VR and 360° video applications. Although these advancements are important for improving history instruction, several serious issues still need to be addressed.

First, information is often presented in text-heavy and cluttered formats, like infographics and banners. These traditional methods lead to low engagement and poor memory retention because they are not interactive and do not allow students to actively participate in understanding prehistoric events.

Second, while digital and web-based interactive timelines have made content easier to access and navigate, they still tend to be two-dimensional. They overlook key aspects of the Visual-Auditory-Kinesthetic (VAK) learning model since they primarily focus on visual exploration rather than auditory or kinesthetic involvement. As a result, despite easier access to chronological information, students often struggle to form lasting connections with the material.

Third, although VR and 360° video methods show potential for presence and immersion, they are often limited to passive viewing. These experiences frequently lack structured teaching elements like gradual reveals, pre-questions, guided narration, and interactive feedback. Without these features, virtual reality remains a tool for visualizing content rather than engaging students in memorable learning experiences.

All these drawbacks highlight the need for an interactive VR timeline system that includes kinesthetic micro-interactions, storytelling, and curiosity-driven elements along with immersion. By using the VAK learning model, integrating storytelling, and empirically evaluating its effectiveness against traditional timelines, the proposed Prehistoric VR Timeline aims to fill these gaps.

1.3 Research Problem

Historically, textbooks, printed timelines, and lectures have been the main static resources used to teach prehistoric ecological evolution. Although these resources offer accurate information, they frequently lack the interactive and immersive elements necessary for a deeper comprehension. Traditional timelines tend to be linear and text-heavy, which lowers student engagement and memory retention, particularly for younger students. Students rarely experience prehistoric events as more than abstract concepts due to the lack of interactive methods, which leaves large gaps in their understanding and enthusiasm for the subject.

Prior studies tried to use 360° video experiences and 2D web-based timelines to enhance the learning of prehistoric subjects. Nevertheless, these approaches mainly made passive observation possible and did not actively engage students in the process of learning. Cognitive science studies highlight how strategies like active recall, gamification, and interactive exploration enhance attention and memory retention. There is a research gap in developing educational tools that integrate immersion and interactivity for learning prehistoric content because current solutions do not adequately apply these principles.

Despite being crucial educational venues, museums and classrooms have limitations when it comes to how they convey prehistoric ideas. Static displays and audio guides are frequently used in exhibits, which offer little opportunity for interaction. Prehistoric studies are less effective as a teaching tool because traditional teaching methods cannot replicate the complexity of ecosystems, animal behaviors, or extinction events. Students lack the resources necessary to directly investigate and engage with these remote eras.

A potent educational tool that connects theory and hands on learning is virtual reality (VR). When compared to conventional teaching methods, research by Freina and Ott (2015) shows how VR improves engagement, focus, and cognitive learning outcomes. Applications of VR-based systems in domains like engineering, medicine, and cultural heritage have demonstrated notable gains in learner motivation and memory retention. The use of VR-based simulations, especially for teaching prehistoric ecological evolution through interactive timelines, however, has not received much attention.

A glaring research issue is the absence of an interactive VR-based timeline that covers prehistoric studies. In addition to being denied immersive settings, learners are also denied cognitive strategies that stimulate curiosity and strengthen long-term memory, such as gamification, guided narration, and pre-questions. Because of this, current teaching strategies fall short in terms of fostering long-term engagement and deeper understanding of prehistoric material.

In order to overcome these constraints, this study suggests creating a Prehistoric VR Timeline. Pre-questions, interactive micro-activities, narrative, immersive 3D

environments, and recap gamification are all integrated into the system. The following research issues are intended to be resolved by this combination

- Absence of Immersion and Interactivity – Present day timelines do not offer significant engagement with prehistoric events, instead they are passive educational resources.
- Short term understanding is the result of traditional teaching methods' failure to implement tried-and-true memory retentive strategies.
- Lack of Practical Exploration – Students are unable to engage in active learning by experiencing extinction events, animal behaviors, or prehistoric ecosystems.
- Underutilization of VR in Prehistoric Education – Although virtual reality (VR) has demonstrated advantages in other domains, its potential for prehistoric ecological education is still not fully understood.

These gaps are filled by the suggested Prehistoric VR Timeline, which turns abstract prehistoric ideas into memorable, captivating, and interactive experiences. The project intends to advance VR-based education while improving learner comprehension, attention, and long-term retention through this method.

1.4 Research Objective

Main objective

The main goal of this study is to create, develop, and test a prehistoric timeline using virtual reality (VR). This timeline aims to provide a memorable, interactive, and multisensory educational experience. Unlike traditional static or digital timelines often used in classrooms, museums, and exhibitions, this system aims to boost learners' attention, curiosity, and memory retention.

Specific Objectives

1. To identify the weaknesses of traditional and digital timeline materials that are used to teach prehistoric events.
 - This involves critically examining current educational materials to spot their weaknesses in engagement, interaction, and retention. Examples of these materials are infographics, posters, and two-dimensional web-based timelines. The findings will reveal the educational gap that the VR system aims to address.
2. To create a virtual reality timeline system that includes kinesthetic micro-interactions, immersive graphics, and narration in the Visual-Auditory-Kinesthetic (VAK) learning model.
 - To ensure inclusivity and better cognitive processing, the design process will focus on combining visual (3D environments and models), auditory (narration and soundscapes), and kinesthetic (controller-based actions and interactions) elements in the VR environment. This approach will meet different learning preferences.
3. To boost student learning and reduce cognitive overload, interactive learning features like progressive reveal (storytelling), guided narration, pre-questions, and responsive feedback should be used.
 - To change passive content delivery into active involvement, we will integrate these elements. Progressive reveal will manage complexity through structured storytelling. Pre-questions will promote focus and

encourage predictions. Responsive feedback will support effective learning practices.

4. To create engaging virtual reality content about the Mesozoic era, we will showcase its three main periods: Triassic, Jurassic, and Cretaceous.
 - Using 3D models and narrated explanations, content development will create realistic ecosystems, species, and evolutionary milestones. The goal is to maintain scientific accuracy and accessibility while providing students with real-world, interactive environments to explore prehistoric life.
5. To compare the learning outcomes of the VR timeline with traditional static timelines, we will conduct experiments to evaluate user engagement, satisfaction, knowledge gain and retention.
 - We use a controlled study design, empirical testing is the main focus of this goal. Static timelines and VR timelines will be used to divide participants into groups. To measure effectiveness and confirm the suggested method, engagement surveys, short-term and delayed retention tests, and usability assessments will be conducted.

2. METHODOLOGY

2.1 Methodology

This study uses a comparative experimental design to assess how effective a Virtual Reality (VR) based prehistoric timeline is compared to traditional static timelines. Participants are split into two groups, one group experiences prehistoric events through a static infographic timeline, while the other group uses the immersive VR system. Both groups are evaluated on memory retention, engagement, and satisfaction through pre-tests, post-tests, and follow-up assessments. This design allows for a controlled comparison of learning outcomes between the two methods. The software development process began with gathering requirements and analyzing feasibility. This was followed by system design, creating 3D content, and designing VR interactions in Unity. The platform was then implemented and improved through pilot testing and user feedback to ensure that both learning objectives and usability were met.

Information sources

This research was conducted through secondary sources, mainly books and documentaries.

- **Books and Academic Literature** - Textbooks and scholarly works on prehistoric life and geological time scales offered clear details about the Mesozoic era, its three periods, and related ecosystems. Educational technology literature helped with the use of VR, interactivity, and the VAK learning model.
- **Documentaries and Multimedia** - Documentaries about prehistoric ecosystems provided visual and narrative clues for rebuilding landscapes, species, and ecological interactions. These helped create realistic 3D assets and engaging storytelling for the VR timeline.

By using books for factual accuracy and documentaries for visual authenticity, the system design provided both scientific reliability and engaging content

Functional requirements

The system must

1. Present prehistoric content based on the three main periods of the Mesozoic era, Triassic, Jurassic, and Cretaceous, using immersive VR environments.
2. Provide a narration that explains the order of events, ecosystems, and key milestones.
3. Allow physical interaction with easy controller inputs, such as teleportation, object selection, and small activities.
4. Provide pre-questions and prompts that spark curiosity before introducing new content to help learners focus.
5. Use gradual reveals and storytelling elements to avoid overwhelming users and keep them engaged. Provide responsive feedback during interactions to reinforce correct understanding
6. Provide a gamified feature (Recap puzzle) to test users memory retention.

Non-functional requirements

The system should

1. Make sure users with little or no prior VR experience can easily navigate using simple controls, like teleport locomotion.
2. Reduce simulator sickness by maintaining steady frame rates, optimizing graphics, and using navigation methods that prioritize comfort. Support usability through clear user interfaces, intuitive feedback, and responsive interactions.
3. Offer the ability to support deployment in schools, museums, and exhibitions.
4. Make sure the system is reliable by stopping crashes and keeping performance smooth on Oculus Quest 3 hardware.

Feasibility study

Technical feasibility

The development of the system will use Unity 3D, Blender for 3D modeling, and luvvoice, Pixabay for Narrations and sound effects. Oculus Quest 3 will serve as the hardware platform, offering enough processing power to support immersive VR scenes with high quality. Current VR development frameworks and optimization methods make the system technically possible.

Operational feasibility

Museums, schools, and exhibitions already use static or digital timelines. Adding VR creates an interactive element that fits well into current educational settings. User-centered design makes sure that learners from various backgrounds can navigate and benefit from the system.

Economic feasibility

The cost of VR hardware, specifically the Oculus Quest 3, is going down. This trend makes the system more accessible for institutions. Most development costs come from software and labor but using open-source or educational licenses helps cut expenses. The modular content design also helps lower long-term costs by allowing reuse and expansion.

Educational feasibility

The combination of narration, VAK learning model strategies, and micro-interactions makes the system effective for teaching. VR improves attention, memory retention, and inclusivity compared to traditional static timelines. This justifies the investment for academic institutions.

System designs

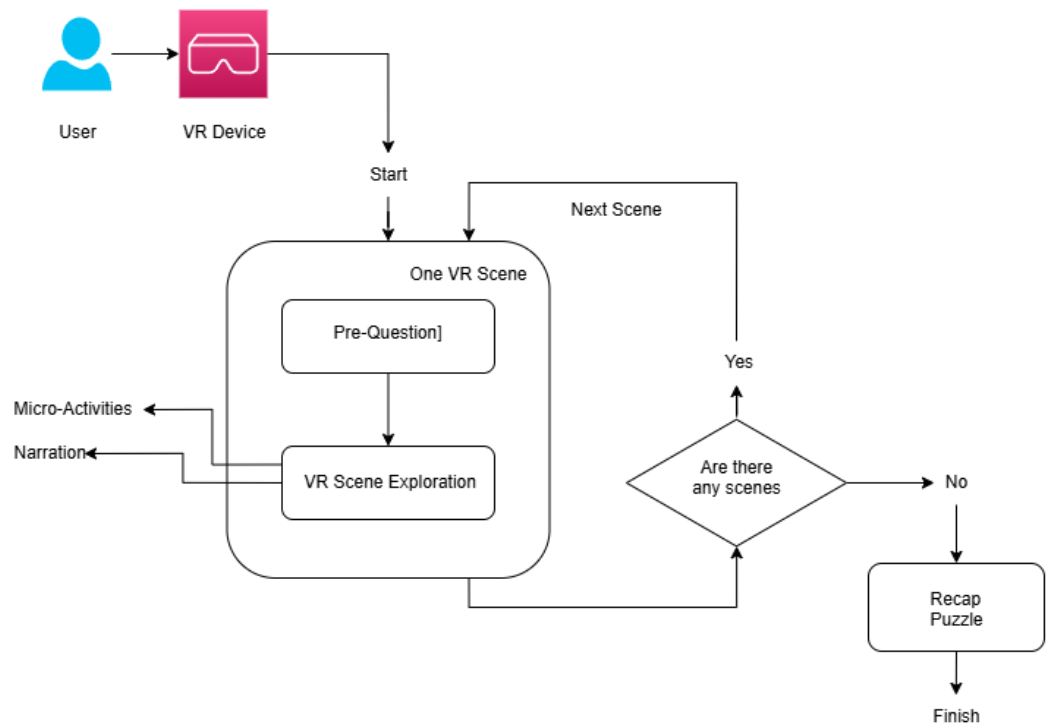


Figure 1 System Overview Diagram

Development professionals used the system design phase to create detailed diagrams of the platform architecture, along with requirement specifications and early prototypes. Key design components include

User interface design

- **Menus and Navigation** - The platform features a simple menu and a teleport-based navigation system. This allows users to move between prehistoric periods without needing any prior VR experience.
- **Interactive 3D Models** - Learners can engage with detailed 3D models of prehistoric plants, environments and animals. This technology offers insights into how evolution changed during the Triassic, Jurassic, and Cretaceous periods.
- **Prehistoric Environments** - Each VR scene recreates landscapes with realistic terrain textures, vegetation, and atmospheric sounds. These immersive elements mimic the ecosystems of the Mesozoic era. They allow students to experience environmental changes over time.

Simulation environment

- **Chronological Flow** - The platform shows events in a sequence, guiding users through the Mesozoic periods and highlighting key evolutionary milestones.
- **Interactive Micro-Activities** - Random activities, like feeding an animal or roam around a specific environment type, keep users from memorizing without understanding and promote active learning.
- **Progressive Storytelling** - Information is shared in stages, helping learners stay curious and engaged without feeling overwhelmed.

The system design showed that it met both technical standards and educational goals. Using immersive prehistoric environments helped learners visualize and understand events better than static media.

Prototyping and iterative development

The validation process through prototyping was essential to confirm the platform's design and functionality. The project followed an iterative development approach. It included regular feedback from educators, and target users at every stage of development. The prototyping phase involved the following steps

Initial Prototypes

- **Low-Fidelity Mockups** - Early concepts were created with wireframes and storyboards. These tools displayed the flow of the timeline and user interactions.

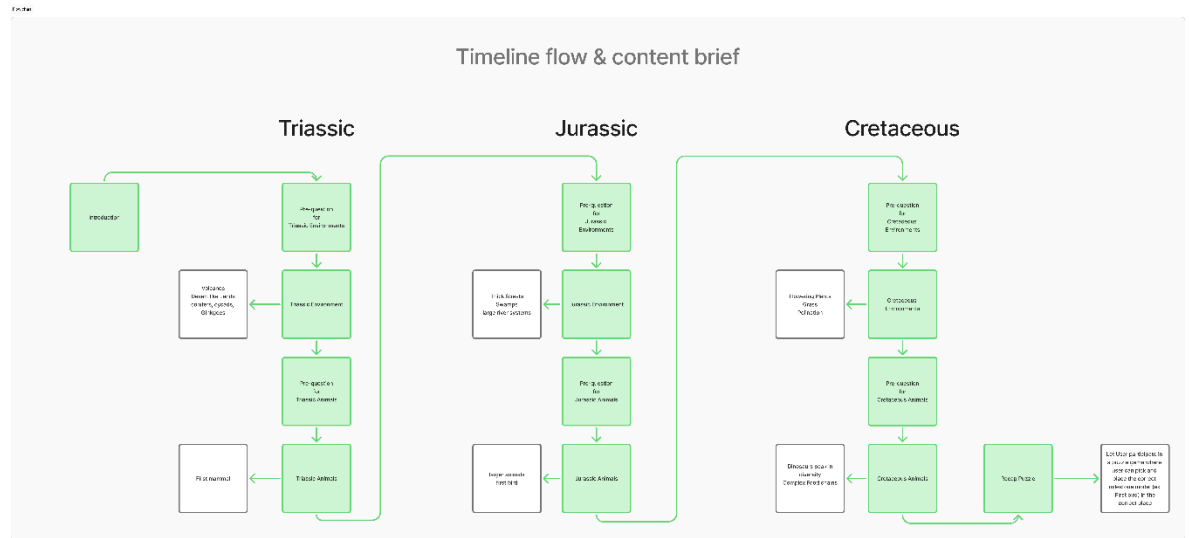


Figure 2 Timeline Flow & Content Brief

- **High-Fidelity Models:** Using premade assets, and the Unity editor to build and combine these models into realistic prehistoric environments.

User Testing

- **Prototype Evaluations** - Sample users, such as students and teachers, tested early versions to assess navigation, clarity of narration, and instructional design.
- **Feedback Collection** - User observations offered information about usability, engagement, and content delivery. This helped the team find weaknesses in the design.

Iterative Refinement

- **Continuous Improvements**- 3D models, animations, and micro-interactions were improved to increase realism, accessibility, and educational value.
- **Feedback Integration**-Recommendations from test users were included to boost interactivity, enhance visual realism, and improve the learning flow.

The platform developed through a series of prototyping cycles that tackled navigation issues, improved user experience, and delivered content better. Ongoing refinement kept it in line with user needs. It also made sure that the system balanced technical

performance with teaching effectiveness.

Technology stack

An educational platform that uses Virtual Reality (VR) needs a mix of modern tools and hardware to provide immersive learning experiences. The Prehistoric VR Timeline uses these technologies.

Virtual reality (VR)

- Hardware - Oculus Quest 3 headsets offer portability and high-quality graphics. They can be used in classrooms, museums, and exhibitions.
- Software - Unity 3D is the main engine for creating interactive simulations. Blender is used for modify 3D models and animation of prehistoric species.
- Interaction- Hand controllers and teleport navigation let learners explore the timeline and interact with objects safely and easily.

3D modeling and animation

- Tools- Blender was used for modify 3d assets.

Each element of the technology stack was chosen to meet both functional needs, such as interactivity, immersion, and accuracy, and non-functional needs, including usability, performance, and scalability.

Content development

Content development used instructional design principles to ensure correctness, engagement, and educational value.

Scenario-Based Learning

- VR environments recreate ecosystems from the Triassic, Jurassic, and Cretaceous periods.
- Learners explore real scenarios like observing changing plants and animals or identifying environmental shifts, which helps them understand better.

Guided learning modules

- Narration gives background on important evolutionary milestones, like the rise of Birds or flowering plants.
- Pre-questions and small interactions help learners predict, observe, and confirm outcomes, improving focus and memory
- Timelines in the VR environment show the order of events and highlight cause-and-effect relationships in how ecosystems evolve.

The content was created together, using sources from paleontology, educational theory, and multimedia storytelling.

User-Centered design principles

The development of the Prehistoric VR Timeline used user-centered design principles to make sure the platform met the needs of its target users, such as students, educators, and museum visitors. These principles shaped the design of the interface, the flow of interactions, and how content was delivered.

Intuitive Navigation

- The system uses clear menu structures and teleport-based movement, which lowers mental effort during exploration
- Consistent design patterns in all modules help learners quickly get used to the environment without needing prior VR experience.

Progressive Learning Pathways

- Modules are organized to help users concentrate on key milestones of the Mesozoic era without feeling overwhelmed by too much detail.
- Progressive reveal techniques make sure information is presented in small, manageable steps. This approach helps avoid overwhelming the mind.

Engagement and Motivation

- Immersive Learning Environment – Realistic prehistoric ecosystems with natural soundscapes improve presence and attention.
- Interactive and Hands-On Learning – Micro-activities like identifying species encourage active participation.

Testing and quality assurance

Testing and quality assurance were done to make sure that the Prehistoric VR Timeline met its functional, non-functional, and educational needs. The process confirmed system reliability, usability, and teaching effectiveness through organized testing phases.

Unit Testing

- Individual components like 3D models, narration modules, and interaction scripts were tested separately.
- Verification confirmed that each module worked properly before system integration.

Integration Testing

- Combined modules, such as narration, interactions, navigation, and feedback systems, were tested to ensure smooth communication.
- We assessed complex situations, like the interactive exploration of ecosystems, for consistency and reliability.

User Acceptance Testing (UAT)

- Students and teachers tested the platform to check its usability, engagement, and educational impact.
- Surveys and observations were used to gather feedback, giving valuable qualitative and quantitative data for ongoing improvement.

Performance Testing

- System responsiveness, frame rates, and memory use were measured on Oculus Quest 3 hardware.
- Stress testing assessed stability during long sessions and many user interactions.

The testing sessions confirmed that the platform is both technically reliable and valuable for education. We made several improvements based on user feedback to make it easier to use. Performance and security tests helped ensure the system is stable, scalable, and safe for use in various learning environments.

Limitations and challenges

Despite the strengths of the Prehistoric VR Timeline, several obstacles and difficulties came up during its development and implementation. Key challenges include

Technical limitations

- The system needs a lot of processing power to create detailed 3D scenes of prehistoric species and ecosystems. It must also keep performance smooth.
- Compatibility issues may happen with older VR hardware. This can limit access in institutions that don't have the latest devices.

Resource constraints

- Budget constraints impacted both the purchase of new hardware and the development of detailed 3D assets.

User adoption challenges

- Some learners and educators were hesitant to use VR technology, especially those who were not familiar with digital tools.
- Orientation and training sessions were needed to help users get used to VR interfaces, navigation, and interaction methods.

Ethical considerations

- The platform needed clear representations of prehistoric species and environments to avoid misrepresentation or misleading information.
- Content design stressed ethical educational practices, promoting respect for extinct life and ecosystems instead of sensationalized portrayals

User testing and iterative development helped tackle these challenges. Orientation programs and user training lowered resistance to adoption. Collaboration with subject experts improved accuracy and teaching quality. Resource limitations were eased by optimizing 3D assets and concentrating on modular design to cut costs.

2.2 Commercialization

Aspects of the Product

A successful commercialization strategy for the Prehistoric VR Timeline will support wide use and a significant educational effect. This section describes the plan for bringing the product to market. It covers Target Audience Segmentation, Market Readiness Assessment, Scalability, Pricing Models, Distribution Channels, Partnerships, and Long-Term Viability.

Target audience segmentation

Educational institutions

- Needs - Schools, universities, and training centers need interactive tools that improve traditional teaching methods and help with knowledge retention.
- Challenges - Limited budgets and infrastructure may slow the adoption of VR solutions.
- Value Proposition - The modular design lets institutions choose specific Mesozoic modules and integrate them into existing curricula for immersive learning.

Museums and exhibitions

- Needs - Museums and exhibitions want engaging, interactive installations to draw in and educate visitors.
- Challenges - Installation and operational costs can be a barrier.
- Value Proposition - The standalone Oculus Quest 3 headset enables easy and low-maintenance setup, making it good for both permanent and traveling exhibitions.

Students and general learners

- Needs - Learners want self-paced, curiosity-driven educational experiences that make prehistory come alive.
- Challenges- Some learners might not have access to VR hardware or may feel unsure about new technology.
- Value Proposition- Interactive storytelling, prediction-based learning, and micro-activities make prehistory engaging and memorable for people of all ages.

Market readiness and scalability

The system uses common VR hardware and popular development tools, like Unity and Blender. This makes it easy for institutions to integrate. The modular design allows for scalability by enabling a gradual rollout, starting with the Mesozoic era and expanding to other prehistoric or historical topics.

Pricing models

- Institutional Licensing - Subscription-based licensing for schools and museums.
- Freemium Model - Basic access for learners with premium upgrades available.
- Educational Discounts - Subsidized pricing for public schools and nonprofit organizations.

Distribution channels

- Direct partnerships with schools, universities, and museums.
- Online distribution through the Meta Quest Store or similar VR marketplaces.
- Work with educational content distributors to reach more people.

Partnerships and collaborations

Partnerships with paleontologists, content creators, and educational designers guarantee accuracy and teaching value. Working with museums and cultural institutions boosts authenticity and helps increase acceptance in public settings.

Long-term viability

The modular system design and flexible pricing models promote long-term sustainability. Regular updates, new content modules, and user-driven improvements will keep it relevant. Ethical standards ensure proper and responsible representation of prehistoric life.

Market readiness assessment

An evaluation of market readiness looked at the demand for the Prehistoric VR Timeline and how well it fits in educational and cultural settings. The assessment included demand analysis, the competitive landscape, technology adoption, and obstacles to implementation.

Demand analysis

- The teaching of prehistoric events still relies on static materials like posters, infographics, and textbooks. These resources often do not engage learners or help them remember long-term.
- The VR timeline fills this gap by offering immersive, interactive, and multisensory learning. It follows proven teaching methods such as the VAK model, gamifications
- Feedback from educators and museum professionals shows a strong interest in VR-based tools that can enhance exhibitions and classroom instruction with active learning experiences.

Competitive landscape

- Current educational programs use traditional methods or, at best, interactive web-based timelines that offer limited immersion.
- Some VR history apps are available, but most concentrate on general historical tours instead of scientifically based prehistoric content.
- The Prehistoric VR Timeline stands out because it combines accurate paleontological content with organized learning methods. This makes it a unique and valuable option in the educational VR sector.

Technological adoption trends

- Falling VR hardware costs and the growth of standalone devices like Oculus Quest 3 make VR easier for schools, universities, and cultural institutions to access.
- Improvements in real-time 3D modeling and interactive storytelling help create immersive educational experiences at lower costs and with better quality.
- More acceptance of VR in education and museums creates a good setting for introducing specialized products like this timeline

Barriers to adoption

- Limited budgets and a lack of VR infrastructure in smaller schools and institutions may slow adoption.
- Some educators and learners may hesitate to use new technology because they are not familiar with VR interfaces.

- Initial implementation needs training sessions for teachers and facilitators to ensure they can use it effectively in classrooms and exhibitions.

The Prehistoric VR Timeline shows great promise in the educational market. It addresses a clear teaching gap, takes advantage of positive trends in VR adoption, and offers a unique product in a less competitive niche. There are challenges like cost and user readiness. However, these can be reduced through modular deployment, training programs, and partnerships with schools and cultural organizations.

Scalability and deployment strategies

The Prehistoric VR Timeline uses scalability features to achieve long-term adoption and widespread educational impact. We created strategic approaches to tackle deployment challenges and make it accessible to institutions, museums, and learners around the world.

Institutional partnerships

- Partnerships with schools, universities, and museums help integrate the VR timeline into curriculums and exhibitions
- Working together allows institutions to share resources, cut costs, and provide steady access to immersive prehistoric education.
- Pilot programs at cultural institutions show their value. They build credibility and support long-term use.

Freemium pricing model

- A basic version of the timeline, covering essential Mesozoic content, can be offered for free to promote access and encourage use.
- Premium features, such as more interactions, extra prehistoric modules, or detailed analytics for educators, would be available through subscriptions or one-time purchases.
- This model balances wide reach with financial sustainability, allowing users to experience the platform before deciding to commit to additional features.

Pricing models

Freemium model

- Basic modules remain free (for example, the introductory Triassic, Jurassic, Cretaceous timeline).
- Premium expansions, like extra content packs or enhanced classroom features, are available through subscription or as one-time purchases.

Institutional licensing

- Depending on the size of the institution, pricing tiers can be scaled to ensure affordability while creating steady revenue streams.
- Schools, colleges, and museums can purchase site-wide or per-student licenses for unlimited access.

Pay-Per-use model

- Because individual users can buy individual sessions or modules, learners without institutional support can use the platform.
- When long-term licensing is impractical, this strategy works well for exhibitions and small-scale deployments.

Distribution channels

Direct Sales to Institutions

- Adoption is strengthened by marketing aimed at museums and schools, bolstered by live demonstrations and pilot installations.
- Training for teachers and facilitators and content customization are made possible by direct relationships.

Workshops and Training Sessions

- Users are introduced to VR navigation and interactions through practical instruction in classrooms and museums.
- Teachers and students gain confidence from step-by-step tutorials, which facilitates easier adoption.

Online Platforms

- Online updates and feature expansions are delivered seamlessly, guaranteeing continued relevance.

- The VR timeline can be listed on educational VR marketplaces like the Meta Quest Store, reaching users worldwide.

Sustainability and long-term viability

Strategic resource planning, varied revenue streams, and continuous enhancements in line with user and institutional requirements are essential to the Prehistoric VR Timeline's long-term viability. Important tactics consist of

Revenue Streams

- The platform is made available to educational institutions, museums, and individual students through pay-per-use models, institutional licensing, and subscriptions.
- Bundled content packages (such as Triassic, Jurassic, and Cretaceous) and possible partnerships with VR hardware suppliers present additional revenue opportunities.
- Diverse revenue streams offer stability in the face of shifting market conditions and lessen reliance on any one model.

Continuous Improvement

- Regular updates, the addition of new prehistoric modules, interactive elements, and enhanced narration help the system stay current.
- Every development cycle incorporates user feedback, which propels improvements in interactivity, content accuracy, and usability.
- In order to sustain high-quality educational experiences, technological developments in VR, AI, and 3D modeling will be continuously incorporated.

Environmental Impact

- By providing digital, reusable, and scalable educational experiences, the VR timeline reduces environmental costs in contrast to physical exhibits that demand significant material resources.

By lowering dependency on printed banners, posters, and throwaway teaching tools, the platform promotes sustainability. This strategy improves its standing in the market by supporting international conservation and sustainability efforts.

Ethical considerations

Strict ethical standards are followed by the Prehistoric VR Timeline to guarantee learner trust, cultural sensitivity, and responsible development.

Cultural Sensitivity

- Content localization guarantees the presentation of prehistoric stories and design components in inclusive and culturally relevant ways.

Educational Integrity

- The materials foster curiosity and respect for Earth's natural history, fostering an appreciation for biodiversity and evolutionary processes.
- The platform keeps free of exaggeration or distortion of prehistory, guaranteeing that content stresses scientific evidence and critical thinking.

Testing devices and tools

Hardware

- VR Headset - The main testing tool was the standalone Oculus Quest 3 headset.
- PC-Based Debugging - Performance optimization, debugging, and profiling were done using Meta Quest Link and a development PC.

Software

- Unity Profiler - To measure CPU/GPU performance, memory usage, and frame rates.
- Google Forms - For collecting structured user feedback during pilot tests.

Testing Environments

- Controlled Lab Testing - To assess system performance, usability, and stability, preliminary assessments were carried out in a laboratory environment.
- Pilot Institutions - To replicate authentic educational environments, testing sessions were set up in the university.

- Field Trials - Additional tests were conducted in exhibition areas under a variety of circumstances, including noisy settings, first-time VR users, and little supervision.

Testers

- Students - Students in schools and colleges with varying degrees of experience with prehistoric material.
- Educators - Testers of usability and content integration for classroom included lecturers.
- General Visitors - To assess accessibility and learning curves for first-time users with little to no prior VR experience were used.

2.3 Testing and Implementation

Testing strategy

The testing procedure was created to make sure the Prehistoric VR Timeline met educational goals, worked properly across all modules, and offered a seamless, engaging experience on the intended hardware. System reliability, usability, and pedagogical value were validated using a systematic approach that combined functional, performance, and user centered testing.

Testing overview and objectives

The main objectives of testing were to

- Confirm that all functional system modules, navigation, narration, and micro-interactions run properly under various hardware scenarios.
- Verify that scene transitions, animations, and user interactions comply to the required levels of responsiveness and realism.
- Compare the platform to static timelines and assess how well it supports memory retention and enhances educational achievement outcomes.
- Find and fix abnormal system behaviors, device-specific performance issues, and usability problems.

Testing methodology and test metrics

Testing phases

Phase 1 – Internal QA (Alpha Testing)

- Conducted by developers and in-house testers.
- Focused on verifying core feature functionality such as narration, teleport navigation, and micro-interactions.
- Evaluated UI navigation flow, performance benchmarks, and detection of major bugs.

Phase 2 – user acceptance testing (UAT)

- Involved real users from three key groups
 - Students
 - Educators
 - General visitors with no prior VR experience.
- This phase emphasized
 - Learning impact (knowledge gain, retention, engagement).
 - Usability and comfort during extended VR sessions.
 - Realism and immersion of 3D prehistoric environments and species.

Functional testing

- Confirmed that all key modules (narrated instructions, interactive micro-activities, 3D prehistoric models, and chronological scene transitions) performed as planned.
- Assessed how responsive interactive elements like object selection/ inspection and environment-based storytelling triggers were.
- Made sure that the Triassic, Jurassic, and Cretaceous evolutionary milestones were accurately depicted in the VR timeline.

Non-Functional testing

- Usability testing - evaluated the clarity of the narration and visual cues, the intuitiveness of the interactions, and the clarity of the navigation.
- Performance Testing - Assessed Oculus Quest 3 hardware frame rates, load times, and scene transition smoothness.
- Reliability Testing - The platform was put through prolonged sessions and numerous user interactions without experiencing any crashes or degradation.
- Accessibility Testing - Confirmed low adoption barriers by demonstrating that new users could finish learning tasks with little assistance.

Implementation plan

Prior to full deployment, the Prehistoric VR Timeline underwent iterative improvements

and small-scale pilot testing. This phased implementation allowed for early problem identification and resolution, ensuring a smoother transition to widespread adoption.

Pilot Testing

- Small groups of students participated in the initial testing.
- Usability, comfort, and educational clarity were the main concerns of early user feedback, which directed enhancements to scene flow, interaction design, and narration pacing.
- User needs like easier navigation and more understandable feedback prompts were identified through pilot session insights and incorporated into the following development cycle.

Iterative Refinement

- The platform's UI navigation, performance stability, and interaction flow were all improved upon in response to pilot testing.
- Immersive storytelling features were improved through successive iterations, which raised learner engagement and authenticity.

Full Deployment

- The system, which offered an engaging educational substitute for static timelines, was made available to larger user groups in educational institutions, museums, and schools.
- The modular design minimized technical disruptions while supporting various deployment contexts, such as public exhibitions and classroom instruction.
- Training workshops and onboarding guides were among the support materials that made adoption easier and guaranteed seamless operation in a variety of learning environments.

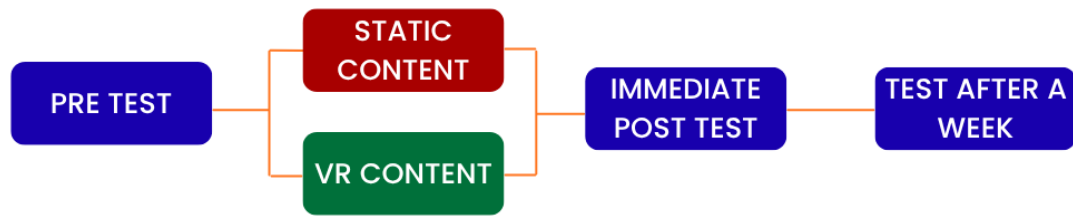


Figure 3 Learning Impact Test Method

The Prehistoric VR Timeline was kept in line with user expectations, technical specifications, and educational objectives thanks to the phased implementation strategy. The platform was able to achieve both operational stability and pedagogical effectiveness through meticulous pilot testing, iterative refinement, and full-scale deployment.

User feedback and iterative improvement

The Prehistoric VR Timeline evolved significantly through structured feedback mechanisms that ensured alignment with educational goals and user needs. Feedback from various user types was gathered, examined, and applied using a methodical approach.

Feedback Collection

- Students participated in usability testing sessions to assess the ease of navigation, narrative clarity, and general efficacy of learning.
- Both quantitative and qualitative information about the advantages and disadvantages of the system was obtained through structured surveys and interviews.
- Feedback indicated useful areas for enhancement, such as content delivery pacing, responsiveness of micro-interactions, and navigation flow.

Iterative Improvement

- User recommendations were incorporated to enhance engagement, improve interactivity, and improve the timeline's educational clarity.

- Based on feedback analytics, improvements were prioritized with an emphasis on the most commonly reported problems, including the accuracy of 3D models, the usability of the interface, and the potency of interactive storytelling.
- Frequent development cycles made sure that updates improved learning outcomes and technical performance over time.

Continuous Feedback Loops

- After every significant update, a continuous system was put in place to gather user input, allowing for adjustment to new specifications and user expectations.
- This feedback was incorporated into periodic updates, which made sure the platform stayed interesting and relevant.
- Constant feedback loops encouraged creativity and adaptability, establishing the Prehistoric VR Timeline as a cutting-edge and competitive teaching resource.

By integrating user-driven enhancements, the platform was able to offer a seamless, captivating, and flexible learning experience in a variety of educational contexts in addition to providing accurate prehistoric content.

Performance metrics

Performance tracking tools were incorporated into the Prehistoric VR Timeline to assess the quality of user interaction and the efficacy of instruction. In order to make sure the platform achieved the desired learning outcomes, quantitative indicators assessed task proficiency, engagement, and knowledge retention.

Task Completion Times

- Micro-activities like relating species to historical periods and engaging with ecosystem components were provided to users.
- The task duration was noted in order to pinpoint areas in which students needed more assistance or more precise teaching cues.
- An examination of completion times showed how rapidly users adjusted to interaction and navigation, offering valuable information about usability and learning effectiveness.

Error Rates

- Errors like incorrect species identification, erroneous event timeline placement, or omitted interaction steps were tracked.

- Real-time feedback was given by error tracking, which highlighted recurring mistakes and prompted remedial advice.
- By helping users identify and steer clear of common errors, this procedure strengthened learning.

Engagement and Interaction Metrics

- We tracked how much time users spent in each scene, how many interactions they finished.
- Scenes with low engagement levels needed to be improved in terms of interactivity, narration, or design.

Knowledge Retention Indicators

- In order to measure content recall and conceptual understanding, pre- and post-assessment results were recorded.
- These metrics demonstrated learning gains and assisted in assessing the efficacy of VAK-based tactics like storytelling, micro-interactions, and narration.

Instructional Adherence

- We looked at whether users interacted with highlighted objects, followed the guided narration, and moved sequentially through the timeline.
- By identifying deviations, the system made sure that students continued to follow the intended course of study.

Performance metrics guided iterative improvements and confirmed the platform's educational impact. The system offered a comprehensive picture of user learning and interaction behavior by integrating task times, error rates, engagement tracking, and retention metrics.

Challenges and limitations

During the Prehistoric VR Timeline's testing and deployment, a number of difficulties and barriers surfaced. To guarantee technical stability, usability, and educational efficacy, these problems needed iterative solutions.

Technical Constraints

- On devices with poor VR performance, rendering high-fidelity 3D models of extinct species and environments occasionally resulted in delays.

- To ensure consistent performance across hardware configurations, optimization techniques like lowering polygon counts and streamlining textures were required.
- A major challenge was striking a balance between system responsiveness and visual realism.

Adaptation Period

- It took some users more time to get used to the navigation and interaction techniques, particularly those who were VR participants for the first time.
- Supervised practice sessions helped learners gain confidence before they explored on their own, addressing initial adoption challenges.

Content Updates

- The platform needed to be updated frequently to maintain scientific relevance and accuracy in light of new paleontology.
- To ensure content validity and instructional alignment, cooperation with educators and subject matter experts was essential.
- To improve usability, storytelling, and interactivity, regular software improvements were required.

Through a combination of technical optimization, user-centered onboarding the Prehistoric VR Timeline overcame these obstacles, guaranteeing its dependability and long-term educational value.

Deployment phases

Every deployment phase used a methodical process that integrated user input and technical development to confirm advancements prior to improvement. Before a full rollout, this phased approach made sure the platform satisfied technical and educational requirements.

System design

- To illustrate the structure and functionality of the platform, the development team created wireframes, architectural diagrams, and prototypes.
- To make things easier for students who have never used virtual reality before, teleport-based navigation systems and user-friendly menus were developed.

- By incorporating user-friendly controls and unambiguous visual cues, the design phase guaranteed usability and engagement, boosting user satisfaction and lowering learning obstacles.

3D modeling and animation

- Detailed 3D models of ancient plants, animals, fossils, and ecosystems were integrated.
- To produce realistic yet performance-efficient visuals, models underwent numerous cycles of optimization and animation.
- Learner engagement and content retention were enhanced by high-quality visuals, which also enhanced immersion.

VR integration

- To ensure seamless operations and expert visual fidelity, core VR components were integrated with the software.
- To make sure the system operated efficiently on Oculus Quest 3 hardware, optimization and compatibility testing were carried out.
- This integration fixed any possible technical issues in various environments while creating an immersive experience.

Narration and interaction modules

- Narrated instructions were incorporated to set the scene for ecosystem transitions and evolutionary milestones.
- To improve clarity and engagement, interactive modules such as species identification, Plant inspection, and micro-activities were improved.
- Feedback loops between interaction and narration made sure that students were focused and engaged with the plot.

Staged deployment

- Alpha testing (internal QA), beta testing (small user groups), pilot testing (Universitas), and full deployment comprised the rollout's phased deployment model.
- Every phase had clear goals and deliverables, and before moving forward, user input was considered.

- The phased approach minimized risks, permitted iterative enhancements, and guaranteed that the platform attained stability and efficacy in education prior to widespread adoption.

3. RESULTS AND DISCUSSION

3.1 Results

The Prehistoric VR Timeline accomplished its intended learning objectives and offered a more engaging substitute for conventional static timelines, according to the findings of the usability tests and educational-effect evaluation. Results are shown in terms of educational efficacy, usability, and user experience.

Usability and user experience

Students, teachers, and general visitors with varying degrees of technical expertise were among the participants. The evaluation was able to capture a wide range of user behaviors and comfort levels because about half of the participants were first-time VR users. All things considered, the system operated reliably, and users needed little help to finish the necessary tasks.

Navigation and interaction

The teleport locomotion system and the simple menu structure were particularly simple to use for the majority of participants. Within the anticipated time, users were able to navigate through the Triassic, Jurassic, and Cretaceous periods of prehistory. Although they needed a brief orientation, first-time VR users quickly adjusted, proving that even novices could use the interaction model. Users were able to move through scenes more easily and with less hesitation thanks to clear visual cues.

Interactive components

Interactive features like the recap puzzle, animal interactions, and plant inspections were often praised as entertaining and inspiring. In contrast to passive observation, participants observed that actions such as identifying species or placing food for the bird created a sense of involvement. By presenting information in manageable steps rather than overloading them with details up front, progressive reveal techniques helped learners stay focused.

User satisfaction

Users consistently reported that the VR experience was more engaging and memorable than infographics or posters across the sample. A greater sense of presence was produced by the integration of spatial audio, dynamic environments, and narration. Teachers noted that compared to traditional materials, the VR format better sustained learners' attention and fostered curiosity.

Adaptation period

Most first-time VR users felt at ease within the first few minutes of interaction, though a few needed assistance at first. As users completed the tasks, observations revealed a consistent decrease in mistakes and irrelevant inputs. The learning curve was lowered by guided orientation and on-screen feedback, which led to high task completion rates overall.

Overall, usability testing verified that the VR timeline was responsive, user-friendly, and appropriate for audiences with a range of technical skill levels. The system was adopted in both educational and public learning settings because it successfully struck a balance between immersion and clarity.

Educational impact

The educational-effect evaluation consisted of pre-tests, immediate post-tests, delayed post-tests, and an engagement survey. In each of the three knowledge tests, the VR group continuously performed better than the static timeline group.

Knowledge gain

Compared to static-timeline learners, who showed an average knowledge gain of 67.1%, VR participants showed an average gain of 88.2%.

This suggests that

- VR almost doubled students' comprehension of Mesozoic events.
- Deeper understanding was aided by active exploration and narration.
- Active recall was strengthened by micro-interactions and prompts based on predictions.

Knowledge retention

By comparing delayed test results with immediate post-test performance, retention was assessed.

- Rate of VR Retention: 71.3%
- 56.2% is the static retention rate.

This demonstrates that a week later, students who used virtual reality retained significantly more information. VR learners maintained a 15.1 percentage-point advantage over static learners despite both groups exhibiting natural score declines.

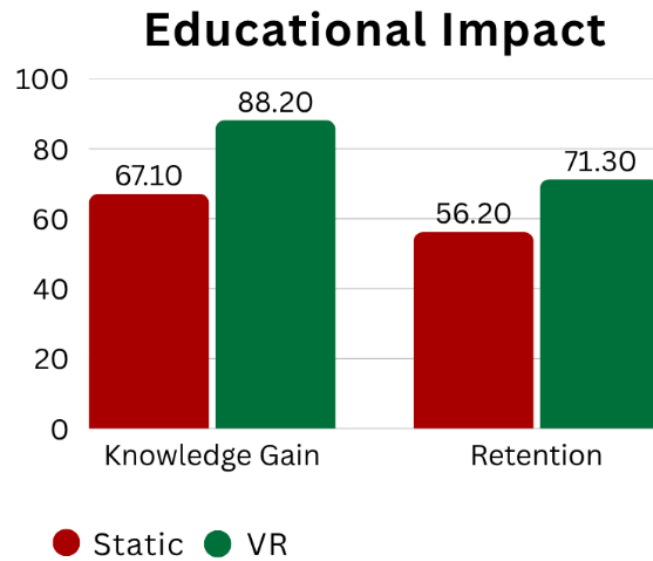


Figure 4 Learning Impact Test Result

Skill mastery

VR participants demonstrated stronger accuracy in

- classification of periods
- Identification of Species
- Identification of ecological milestones

Interactive activities greatly decreased misconceptions and encouraged better decision-making, particularly the recap puzzle. In general, VR outperformed static learning techniques in terms of improving conceptual understanding, strengthening long-term retention, and improving immediate learning.

3.2 Research Finding

Comparative analysis of traditional methods

To assess its educational impact, the Prehistoric VR Timeline was contrasted with traditional teaching techniques like textbooks, and static infographic displays. The analysis showed that VR-based learning has distinct advantages over conventional methods.

Engagement levels

When using VR, participants consistently reported higher levels of engagement. Compared to posters or printed diagrams, the immersive scenes, animated species, and hands-on interactions helped maintain attention for much longer. Students said the virtual reality experience was more "real," "memorable," and "easier to understand."

Learning outcomes

Metric	Static Timeline	VR Timeline
Knowledge Gain	67.1%	88.2%
Retention Rate	56.2%	71.3%
Chronological Understanding	Moderate	High
Species Identification Accuracy	Moderate	High

Table 1 Learning Outcome

In all three testing phases, VR learners continuously scored higher. They were able to develop stronger mental connections through the use of narration, environmental immersion, and interactive tasks, which improved their short-term comprehension and long-term memory.

Safety and accessibility

- In contrast to physical exhibits or materials, the VR timeline offered scalable accessibility, allowing a variety of audiences from schoolchildren to museum visitors to explore prehistoric content regardless of prior knowledge.
- The VR system offered a risk-free environment, allowing learners to experiment, make mistakes, and repeat activities without penalty.

Challenges identified

Despite achieving its objectives, the Prehistoric VR Timeline encountered a number of difficulties throughout its creation, testing, and implementation. These challenges offer crucial guidance for enhancing subsequent iterations of the system.

Technical constraints

- On low-end VR devices, high-fidelity 3D models of extinct animals and environments occasionally caused performance lag, necessitating optimization techniques like lowering polygon counts and streamlining textures.
- A major obstacle across various hardware configurations was ensuring realistic frame rates.

Content updates

- As new paleontological findings and interpretations are made, the platform needs to be updated frequently to maintain its scientific accuracy.
- Regular software updates and ongoing cooperation with subject matter experts are required to maintain accuracy and educational value.

User adaptation

- First time users required extra assistance to get over their inexperience with immersive interfaces, even though many participants adjusted to the VR environment quickly.
- Users gained comfort and confidence through guided tutorials, and system based problem solving prompts however, learner groups' adaptation times still differed.

The difficulties noted point to areas that require constant development, such as technical optimization, expert cooperation for content updates, and improved onboarding materials to better assist new VR users.

3.3 Discussion

The comparative evaluation's results clearly show that the Prehistoric VR Timeline has a number of advantages over conventional static learning resources. Immersion settings, interactive micro-activities, and guided narration all worked together to produce quantifiable gains in learner engagement and comprehension.

The difference in knowledge gain between the two groups is one of the study's main findings. VR participants outperformed static-timeline learners, who demonstrated a 67.1% gain, with an average improvement of 88.2%. The added value of interactive and multisensory learning is reflected in this nearly 21 percentage point advantage. Learners developed deeper conceptual understanding as a result of the spatial arrangement of prehistoric environments, narration, and prediction-based prompts.

This conclusion is further supported by the analysis of long-term retention. VR participants retained 71.3% of their knowledge a week after the learning session, while the static group only retained 56.2%. This is a significant improvement of 15.1 percentage points, suggesting that VR facilitated longer-lasting learning. The increased retention implies that interaction-driven recall, like finishing the recap puzzle or engaging in ecosystem-based activities, stabilized memory traces more effectively than passively viewing static content.

These quantitative results were supported by qualitative comments. Compared to infographics or textbook diagrams, participants often said that the VR experience was more "memorable" and "clear." Learners were able to process events chronologically rather than as discrete facts thanks to the sense of presence that 3D environments and natural spatial audio created. This seemed to lessen cognitive overload. Teachers also noticed that students who used virtual reality (VR) were more comfortable describing evolutionary transitions than students who used traditional timelines.

But there were a few small difficulties with the switch to VR. A few novice VR users needed some initial assistance to grasp controller input and navigation. After a quick orientation, these problems quickly subsided, and no notable usability obstacles were noted. Crucially, motion discomfort stayed minimal, satisfying suggested comfort standards for brief instructional virtual reality sessions.

Overall, the conversation emphasizes how immersive VR improves both long-term memory consolidation and short-term comprehension. The findings demonstrate that compared to static timelines, the Prehistoric VR Timeline offers a more comprehensive, captivating, and successful educational experience. The system supports a variety of learning styles and provides an engaging substitute for classrooms, museums, and public education settings by incorporating storytelling, interaction, and multisensory cues.

4. CONCLUSION

This study shows how traditional static, text-heavy timelines can be transformed into immersive, interactive learning environments using the Prehistoric VR Timeline. The system combined interactive 3D models, scenario-based learning activities, and narrated guidance to make prehistoric content more memorable and engaging. The project effectively achieved its main goals of enhancing learner confidence, comprehension, and long-term retention by using this strategy.

Requirements collection, feasibility analysis, iterative prototyping, and ongoing user-based improvement were all part of the systematic process used to develop the platform. Feedback from educators, students, and first-time VR users played a key role in shaping the system's usability and instructional design. Throughout the project cycle, both scientific accuracy and user experience were preserved thanks to this iterative workflow.

VR clearly outperformed static timelines in terms of instruction, according to the evaluation. VR participants showed an 88.2% gain and a 71.3% retention rate, compared to static learners' 67.1% knowledge gain and 56.2% retention rate. Stronger chronological comprehension, improved species identification, and increased confidence in the explanation of evolutionary transitions were further indications of these advancements. Immersion settings, spatial narration, and micro-interactions all worked together to support deep learning as opposed to surface-level recall by converting abstract information into tangible experiences.

Technically speaking, the project demonstrated the feasibility of utilizing Blender to refine 3D assets and Unity as the primary development engine. Without the need for external computers, the Oculus Quest 3's reliable and easily accessible hardware platform allowed for seamless standalone deployment. While onboarding tutorials assisted new VR users in quickly adjusting to the system, optimization techniques made sure that performance remained constant.

The VR timeline's modular design allows for future improvements, such as more prehistoric eras, more ecological interactions, or more historical content. It is also a useful tool for public exhibitions, museums, and classrooms due to its stand-alone nature.

All things considered, the Prehistoric VR Timeline shows how effective immersive VR can be as a teaching tool. The system provides a cutting-edge, successful, and captivating substitute for conventional teaching techniques by fusing interaction, narrative, and scientifically accurate settings. It can be further developed and modified to promote educational innovation in the future.

5. REFERENCES

- [1] H. Zhang, Y. Wang, G. Fan, Y. Song, and Y. Hu, “Multimodal teaching analytics: the application of SCORM courseware technology integrating 360-degree panoramic VR in historical courses,” *Scientific Reports*, vol. 13, no. 18893, 2023, doi: 10.1038/s41598-023-46229-2.
- [2] A. S. Deininger and R. Halim, “Online interactive timeline to improve learning of history in school education,” *Proceedings of the International Conference on Computer Applications & Information Security*, 2021.
- [3] P. P. Nguyen, “Exploring historical events through an interactive timeline,” *ProQuest Dissertations Publishing*, 2020.
- [4] J. Woodward (Ed.), *Knowledge Encyclopedia: Dinosaur! (Revised and Updated Edition)*. DK Publishing, London, 2022.
- [5] N. Fleming, *Teaching and Learning Styles: VARK Strategies*. Christchurch, New Zealand: N.D. Fleming, 2001.
- [6] J. Jerald, *The VR Book: Human-Centered Design for Virtual Reality*. Morgan & Claypool, 2015.

6. APPENDICES



Figure 5 Appendix 1



Figure 6 Appendix 2



Figure 7 Appendix 3

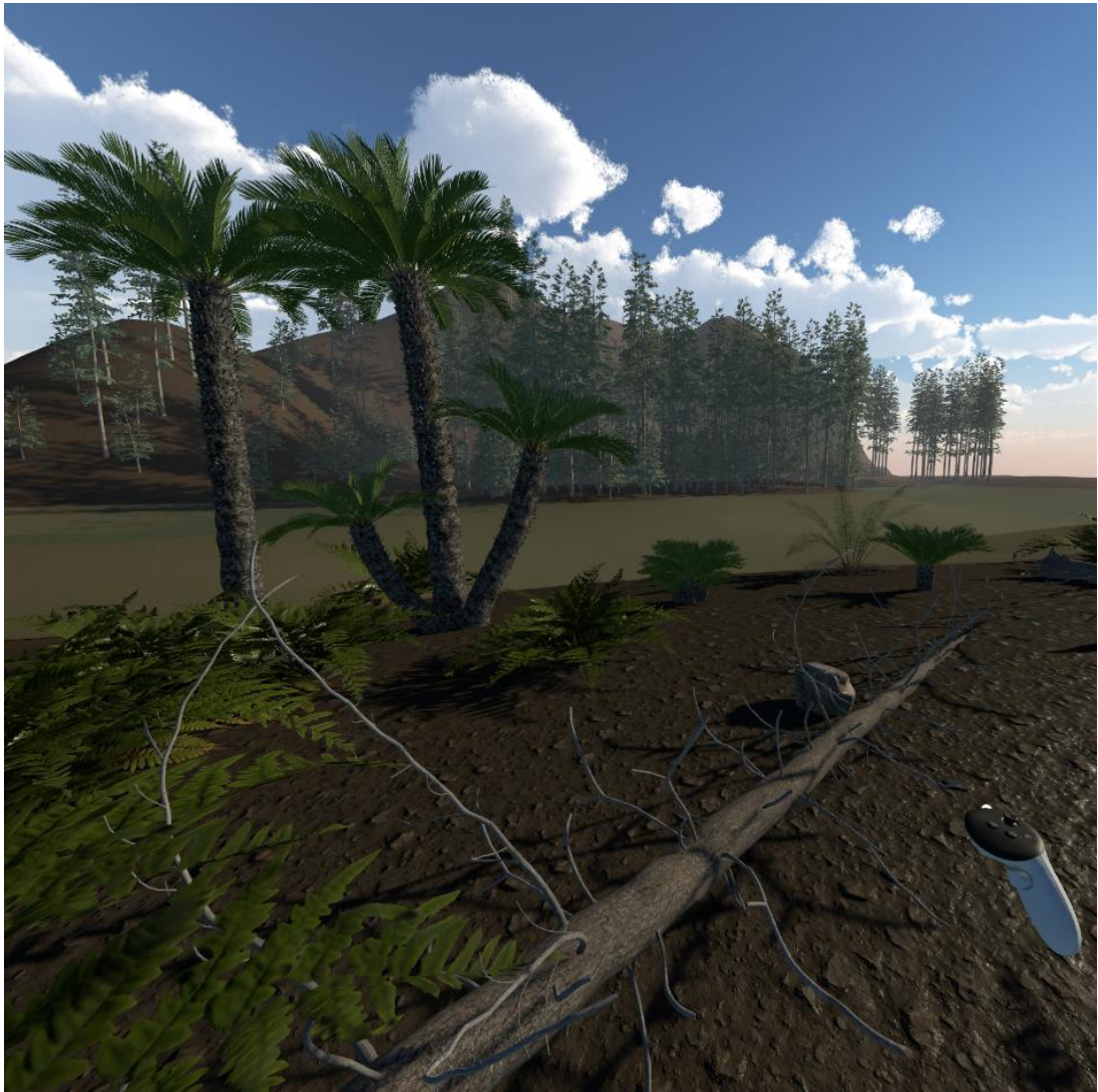


Figure 8 Appendix 4



Figure 9 Appendix 5



Figure 11 Appendix 7